Development of Composite Clinical-Radiological Tool to Predict Functional Outcomes after Ischemic Stroke Treatment

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ABSTRACT

Quantitative radiological tools to assess acute ischemic stroke (AIS) survivor's functional outcomes are limited. While conventional qualitative scoring systems exist, they are limited by inter-rater variability due to subtle ischemic changes. Post-intervention estimation of final cerebral infarct volume (CIV) followed by assessment is a known objective radiological determinant of functional outcomes. Hence, this study aims to develop and evaluate a composite radiological tool using radiomic imaging markers to predict long-term functional risk in AIS survivors. The dataset consists of 50 AIS patients' clinical and radiological information. First, Alberta stroke programme early CT score (ASPECTS) and posterior circulation-ASPECTS regions were annotated on scans followed by mapping them with CIV. Multiple volumetric parameters were extracted, including TBV, CIV, and the proportion of CIV to TBV. These raw features were used to compute percentage volumes and perform summation and proportion analyses w.r.to ASPECTS regions. Premorbid and clinical features were also converted to meaningful representations. Principal Component Analysis and Recursive Feature Elimination (RFE) were employed to identify optimal feature sets. Finally, different combinations of composite features were utilized to train classification algorithms. Promising results were achieved with RFE using both feature combination approaches. Support Vector Machine (SVM) on raw features achieved the highest AUC value (0.94±0.05), while Logistic Regression (LR) attained AUC value (0.93±0.05). Additional analysis indicated models trained using transformed features perform more consistently and achieve superior overall performance. The study demonstrated the potential of composite tool using radiomic and clinical information to accurately predict AIS patient's risk of long-term functional outcomes.

Keywords: Ischemic Stroke, ASPECTS, Quantitative Approach, Radiomics, Volumetric Analysis, Imaging Markers, Data Science, Machine Learning

INTRODUCTION

Acute ischemic stroke (AIS), caused by blood clots obstructing brain arteries, leads to reduced blood flow and oxygen, resulting in severe neurological damage or death. In 2022, over 7.6 million new AIS cases were reported globally, accounting for more than 62% of all strokes. Approximately 3.3 million of these cases resulted in death, with stroke-related fatalities and disabilities contributing to over 63 million disability-adjusted life-years (DALYs) annually[1], [2]. The prognoses for patients with AIS are primarily determined using subjective clinical scales [3], [4] and depend on the extent of brain tissue damage [5]. A robust prognostic model is vital for assessing treatment effects, follow-up care, and making timely adjustments to the patient's social environment, such as arranging early long-term care [6]. Alberta Stroke Program Early CT Score (ASPECTS)/AP is a scoring system initially developed to assess early ischemic changes on pre-intervention non-contrast brain CT scan that subdivides anterior circulation brain territories into 10 different regions [7]. *AP* provides a systematic and practical method for evaluating early ischemic changes, offering better reliability and interobserver agreement compared to the one-third middle cerebral artery (MCA) rule, which requires clinicians to estimate if more than one-third of the middle cerebral artery (MCA) territory is affected by ischemia on a CT scan [8]. This method was extended to include the posterior circulation territories, known as PC-ASPECTS (*PA*) [9]. The 10-point topographic system is commonly utilized in pre-intervention scans to guide clinical treatment decisions[10] and is also recognized as a prognostic marker for functional outcomes following intervention [11], [12]. MRI scans have high sensitivity in evaluating post-intervention cerebral infarct volume (CIV) which is a well-established predictor of future outcomes in AIS patients [13], [14]. Incorporating information from *AP/PA* regions and subdividing the CIV can provide more granularity in outcome assessments. Artificial intelligence (AI) and machine learning (ML) advancements are highly reflected in improving stroke diagnosis and prognosis assessment. Studies suggest that implementing these technologies

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using automatic extraction and analysis of features from imaging scans can significantly enhance the precision of stroke-related assessments[15] and can overcome the limitations of observer variability.

This pilot study assesses the feasibility of developing an optimal prognostic model to predict functional outcomes of patients using a comparative volumetric analysis with multiple machine learning algorithms. The study evaluates infarcted tissues across 20 regions defined by *AP* and *PA*. Specifically, it examines the ratio of CIV in *AP* and *PA* regions, derived from post-intervention MRI and CT scans (performed ~24hrs of AIS onset), relative to the total brain volume (TBV), treating these ratios as potential imaging markers. These markers are analyzed in combination with pertinent clinical characteristics of the patients. Notably, this pilot study is novel and aims to develop and evaluate a composite radiological tool using radiomic imaging markers to predict long-term functional risk in AIS survivors.

METHODOLOGY

A. Data Collection:

The Institutional Review Boards of UT Southwestern (UTSW) and the University of Oklahoma (OU) approved the retrospective study. The patient inclusion criteria required confirmed AIS and thrombolysis treatment at Clements University Hospital and Texas Health Presbyterian Hospital between 2019 and 2022. Study protocol design, data collection, and de-identification were conducted at UTSW whereas, OU conducted analysis and model development. The dataset consisted of fifty patients with clinical and radiological information collected during the assessment period.

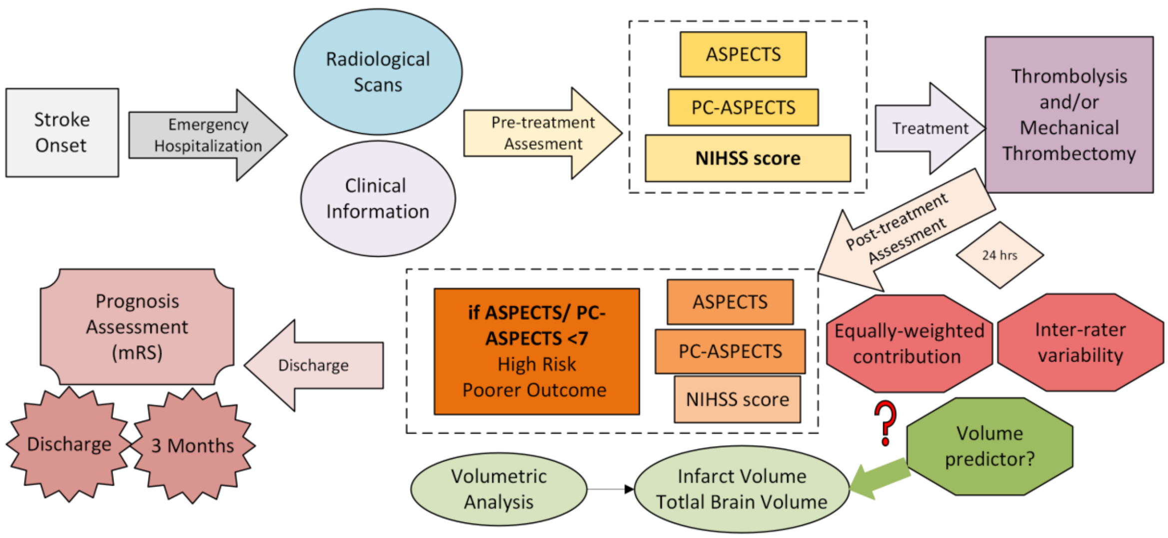

Figure 1: Traditional prognosis assessment and sustaining limitations.

The motive of this study and the limitations in current protocols for assessing ischemic stroke patients are presented in Figure 1. It outlines a typical patient's treatment protocol from the initial symptom onset; screening methods; treatment procedures; and prognostic assessment. *AP*/*PA* score and the National Institutes of Health Stroke Scale (NIHSS) conducted pre- and post-treatment, which guides doctors in treatment planning and assessing potential outcomes. Modified Rankin Scale (mRS), to evaluate patients' degree of dependency is also collected at discharge and the 3-month follow-up. Even though multiple radiological examinations were acquired, only post-treatment scans were used in our analysis. Due to the qualitative nature of the *AP/PA* assessment, it is still unclear whether each of these regions should be weighed equally, or whether quantitative assessment of the CIV volumes with respect to the *AP/PA* regions would be an ideal composite tool for prognostic assessment.

B. Data Pre-processing & Feature Extraction:

i. Imaging Data:

MRI sequences (T2/FLAIR) were primarily used for *AP/PA* annotation due to their superior anatomical resolution, while DWI/ADC was used to mark CIV. Expert doctors carefully examined each case using ITK-SNAP to perform CIV ground truth marking. We developed an in-house CAD system based on our previous studies [16], [17] to facilitate the rest of the capabilities such as automated TBV extraction, *AP/PA* region annotation, cross-referencing, and validation of discrepancies. CT scans were used for feature extraction when MRI scans were unavailable. Ten anterior regions (from *AP*) were marked: *M1*, *M2*, *M3*, *M4*, *M5*, *M6*, insula (*I*), internal capsule (*IC*), lentiform (*L*), and caudate (*C*) [8]. Additionally, ten posterior regions (from *PA*) were annotated, including the left and right sections of the midbrain, thalamus, occipital lobes, pons, and cerebellar hemisphere [9]. All annotated scans were then mapped to infarct volume data. The number of slices per case ranged from 30 to 36 with average brain and infarct volumes of 1312.63 ± 136.63 mm^3 and 60.01 mm^3 ± 77.38 mm^3 respectively.

ii. Clinical Data:

Several key indicators were collected to assess the patient's condition including demographics, health profile, pre-stroke conditions, blood tests, and stroke manifestation. Specifically, demographics included age and gender; existing health profile indicators were BMI, diabetes, and tobacco usage. Tests were conducted to document diabetes status (HbA1C and glycemic gap), hyperlipidemia, and atrial fibrillation. Regarding stroke manifestation, etiology, time to thrombolysis, and NIHSS at both admission and 24h were also collected. Patients' mRS at both discharge and 3 months were evaluated to comprehensively assess patient recovery and treatment efficacy.

C. Feature Engineering:

The raw features from the above extraction phase underwent required transformations to appropriately capture and represent clinical and radiomic information. For instance, NIHSS was categorized into 5 scales (0 to 4) representing non-minimal (0-4), mild (5-9), moderate (10-14), moderately severe (15-19), and severe (>20) categories respectively [18]. Time to thrombolysis was also categorized into three scales (0 to 2) [19], highlighting fast (55-129 mins), delayed (130-204 mins), and extremely delayed (205-280 mins) responses respectively.

The proposed CAD system computes quantitative features related to *TBV*, *CIV*, and *AP/PA* raw volumes (20 values). It also computes mapped volumes (20 values), which represent the overlap/intersection of volumes between *CIV* and *AP/PA* regions. These features were used to construct our first feature set combination (FS_1). Additional transformations were done with the second feature set (FS_2) and were applied to the mapped volumes to create individual proportional volumes (VP_i); sectional volumes VS_{AP}, VS_{PA}, VS_T, w.r.to *AP*, *PA*, and both regions respectively. Finally, the presence of *CIV* in *AP/PA* regions is also created as a binary flag. More information about the two feature sets (FS_1, FS_2) as well as the transformations were presented in Table 1. Extracted clinical features were also standardized to ensure compatibility with the modeling process. The target variable (mRS) has also been converted into a binary representation, where 0 means good prognosis (negative class) (mRS: 0-2), and 1 means poor prognosis (positive class) (mRS: 3-6) outcomes.

Table 1: Details about the proposed feature set and transformations

Feature Set	**Features and/or transformations**
FS_1	All features from clinical and radiomic transformation
FS_2	All features from clinical transformation + additional transformations listed here: $[1].\ \mathrm{VP_i} = {}^{\text{map volume}_i}/_{\mathrm{CIV}};\ \ [5].\ \sum \text{if } (\text{map volume}_i) > 0$ $[2,3,4].\ \mathrm{VS_{Type}} = {}^{\sum_{i=x\to y} \text{map volume}_i}/_{\mathrm{CIV}},\ \text{if } \begin{matrix} \text{Type} = \text{AP};\ x=1, y=10 \\ \text{Type} = \text{PA};\ x=11, y=20 \\ \text{Type} = \text{T};\ x=1, y=20 \end{matrix}$

D. Proposed Model Design and Evaluation:

The two feature sets were used to develop several classification models to assess good and poor prognoses. Each model generates classification probability scores ranging from 0 to 1. The higher score represents the higher likelihood of the case having a poor prognosis (positive class: mRS = 3-6). To build multi-feature fusion classification models, we selected four well-known architectures namely: Support Vector Machine (*SVM*), Random Forest (*RF*), Logistic Regression (*LR*), and Extreme Gradient Boosting (XGB). During optimal model building, several key steps were considered, first, feature-wise standardization is applied to transform each attribute. Second, three combinations for each of the two feature sets were explored including (a) raw features; (b) feature dimensionality reduction with principal component analysis (*PCA*) with a variance rate of 95%; (c) feature selection approach using recursive feature elimination (*RFE*). Third, to address the small dataset, a leave-one-case-out (*LOCO*) based cross-validation method is adopted to train and evaluate each ML model to maximize the number of training samples and avoid case partition bias [20]. Finally, to address the class imbalance, during each step of LOCO training, the synthetic minority over-sampling technique (*SMOTE*) [21] was applied to create synthetic samples in the minority class using the true sample distribution allowing for more robust training and validation. Figure 2 shows the flow diagram of each step in our CAD system including data preparation, transformation, optimal model design, and evaluation.

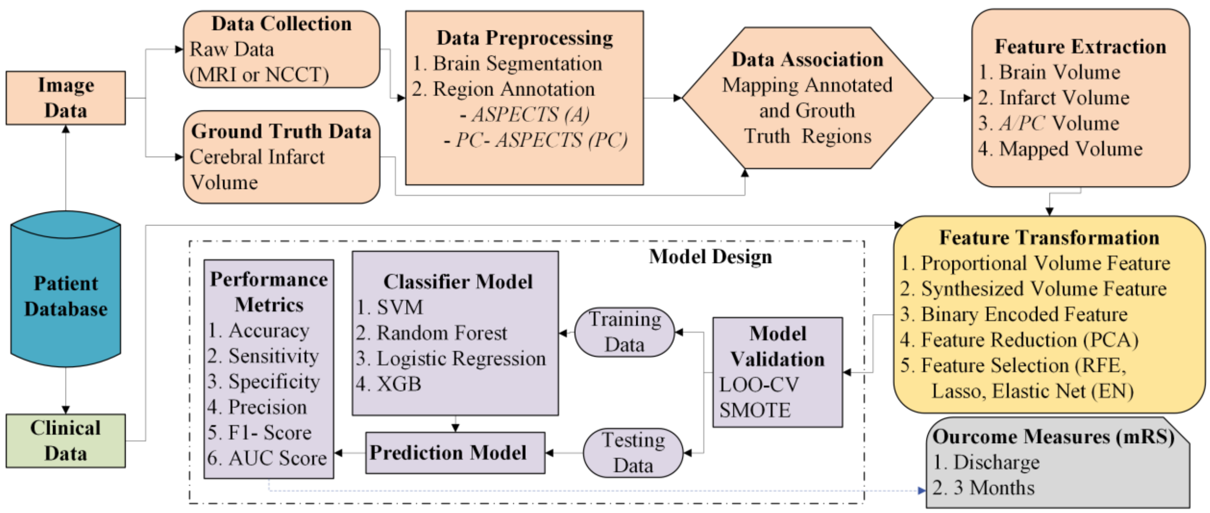

Figure 2: The framework of the Prognostic Model for Ischemic Stroke Assessment.

To evaluate the performance of each classification model, we used two approaches. First, a receiver operating characteristic curve (ROC) is constructed using classification scores, and the area under the ROC curve (AUC) is computed to assess, evaluate, and compare the performance to classify the mRS discharge. Second, an operating threshold (T = 0.5) was used on classification scores to divide all testing cases into two mRS classes (score $\leq$ 0.5: good prognosis; score > 0.5: poor prognosis). Classification results were then used to compute several performance metrics (i.e., accuracy, precision, sensitivity, and specificity).

RESULTS

Error! Reference source not found. compares the performance metrics of several classifier models using two feature sets and three combinations defined above. Using FS_1, the accuracy ranged between 0.52 and 0.91, with the highest score for *SVM-RFE*. It can also be noted that *PCA* has no effect, however, *RFE* has contributed to improved performance. Further, *SVM-RFE* recorded optimal performance for precision (0.86), sensitivity (0.86), and specificity (0.94). Next, using FS_2, the accuracy ranged between 0.54 and 0.89 with the *LR-RFE* achieving the highest value. Similar to FS_1, FS_2 also showed the addition of *PCA* for dimensionality reduction having a negative effect, while *RFE* generally has a positive impact. Overall, both *LR* and *SVM* with *RFE* had the most balanced results with *LR* being significantly better. Overall, FS_1 tends to be a better pick compared to FS_2. Also, *LR* and SVM with RFE performed significantly better than other approaches, while *RF* and *XGB* variations performed the worst.

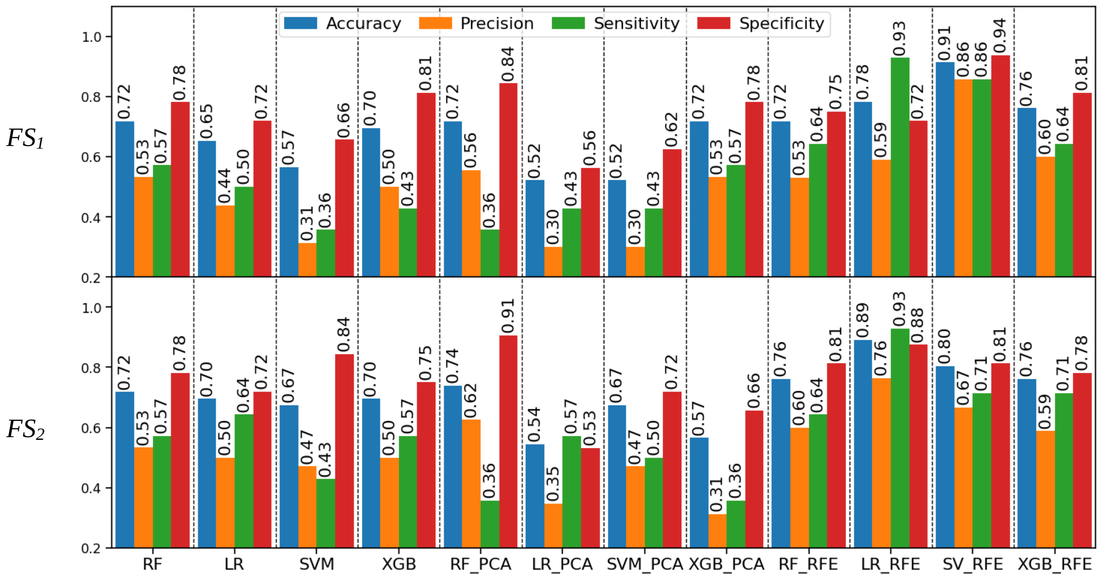

Figure 3: Comparison of performance between approaches; top: FS_1; bottom: FS_2.

Furthermore, Figure 4 presents the ROC curves analyzed using AUC scores. *FS_1* still has the highest AUC score of 0.94 using the *SVM-RFE*, the second best reported for *LR-RFE*. Similarly, for *FS_2*, *LR-RFE* and *SVM-RFE* performed the best with 0.92 and 0.85 AUC values respectively. In summary, the *RFE* approach yielded the best results, whereas *PCA* has the worst performance. Both *SVM* and *LR* using the *RFE* approach yielded significantly better results, with *FS_1* being a slightly better choice compared to *FS_2*.

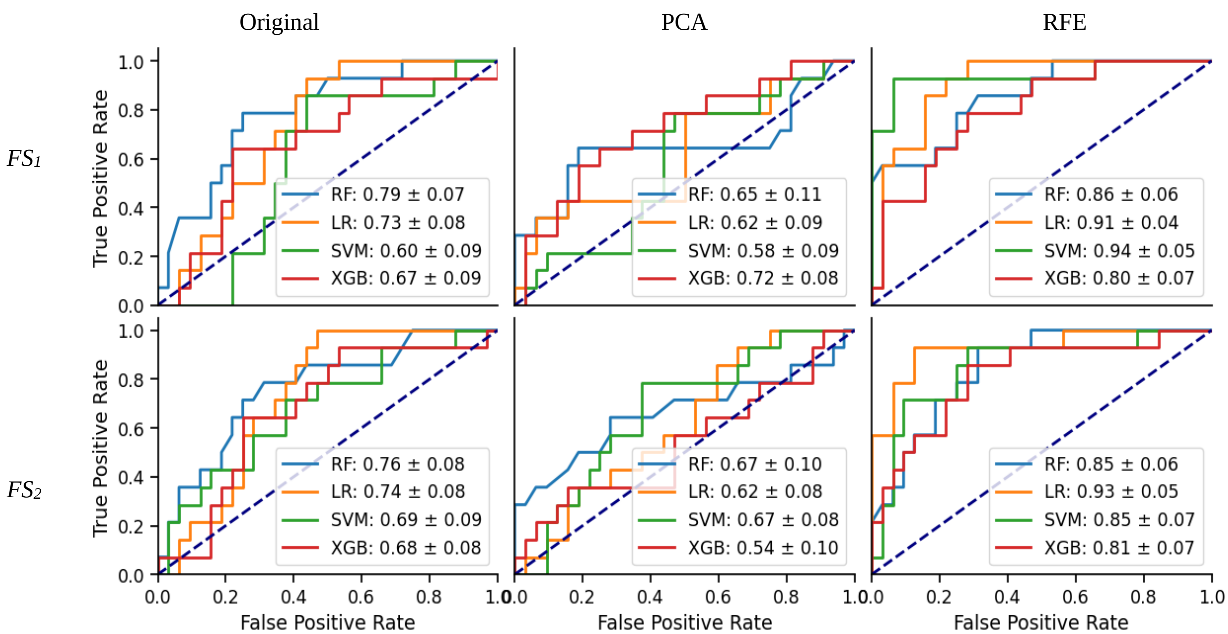

Figure 4 Comparison of AUC curves between two approaches using three different combinations.

DISCUSSION AND CONCLUSION

The traditional approaches for AIS stroke prognostic prediction have several limitations including qualitative assessments, not considering the extent of AP/*PA* region's effect of CIV, and inter-rater observer variability [22], [23]. To address these shortcomings, this study explores the use of a quantitative approach, integrating volumetric information from clinically relevant regions with clinical attributes, to develop a composite tool to predict long-term functional outcomes/risks in AIS survivors following treatment. The study has several significant research contributions to better understand risk prognosis in AIS survivors. First, it demonstrates the effectiveness of quantitative features in *AP/PA* regions for predicting functional outcomes. High AUC scores indicate the model's robust ability to differentiate between favorable and unfavorable prognoses. Second, comprehensive extraction and analysis of radiomic features combing different feature engineering and transformations were conducted and showed promise in assessing their significance. **Error! Reference source not found.** and Figure 4 illustrate the comparison of performance metrics and AUC scores between models utilizing raw and transformed features, highlighting the enhanced predictive capability of the quantitative approach. Lastly, the study employed and compared multiple classification algorithms to identify the optimized prognostic model, further validating the approach's efficacy.

In conclusion, the preliminary results on the limited dataset highlight the study's potential in predicting functional outcomes (mRS at discharge) in AIS survivors. We have plans to expand the study population significantly (~200) patients and include long-term outcomes (mRS at 3 months) in our prospective studies. The future work will also validate the research findings on the larger and diverse dataset, perform extended analysis, and explore and compare the potential of several subsets of feature subsets.

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REFERENCES

[1] V. L. Feigin *et al.*, “World Stroke Organization (WSO): Global Stroke Fact Sheet 2022,” Jan. 01, 2022, *SAGE Publications Inc.* doi: 10.1177/17474930211065917.

[2] G. Oliveira, A. C. Fonseca, J. Ferro, and A. L. Oliveira, “Deep Learning-Based Extraction of Biomarkers for the Prediction of the Functional Outcome of Ischemic Stroke Patients,” *Diagnostics*, vol. 13, no. 24, Dec. 2023, doi: 10.3390/diagnostics13243604.

[3] C. D. Nouh *et al.*, “Quantitative Analysis of Stress-Induced Hyperglycemia and Intracranial Blood Volumes for Predicting Mortality After Intracerebral Hemorrhage,” *Transl Stroke Res*, vol. 13, no. 4, pp. 595–603, Aug. 2022, doi: 10.1007/s12975-022-00985-x.

[4] K. P. O’Connor *et al.*, “Predicting Clinical Outcome After Mechanical Thrombectomy: The GADIS (Gender, Age, Diabetes Mellitus History, Infarct Volume, and Sex) Score,” *World Neurosurg*, vol. 134, pp. e1130–e1142, Feb. 2020, doi: 10.1016/j.wneu.2019.11.127.

[5] V. Dumas *et al.*, “Functional outcome in low-ASPECTS (0–5) acute ischemic stroke treated with mechanical thrombectomy: impact of laterality explored in a single-center study,” *Front Neurol*, vol. 14, 2023, doi: 10.3389/fneur.2023.1205256.

[6] H. C. Kniep *et al.*, “Imaging-based outcome prediction in posterior circulation stroke,” *J Neurol*, vol. 269, no. 7, pp. 3800–3809, Jul. 2022, doi: 10.1007/s00415-022-11010-4.

[7] P. A. Barber, A. M. Demchuk, J. Zhang, and A. M. Buchan, “Validity and reliability of a quantitative computed tomography score in predicting outcome of hyperacute stroke before thrombolytic therapy,” *Lancet*, vol. 355, no. 9216, pp. 1670–1674, May 2000, doi: 10.1016/S0140-6736(00)02237-6.

[8] J. H. Warwick Pexman *et al.*, “Use of the Alberta Stroke Program Early CT Score (ASPECTS) for Assessing CT Scans in Patients with Acute Stroke,” 2001.

[9] H. Tei, S. Uchiyama, T. Usui, and K. Ohara, “Posterior circulation ASPECTS on diffusion-weighted MRI can be a powerful marker for predicting functional outcom,” *J Neurol*, vol. 257, no. 5, pp. 767–773, May 2010, doi: 10.1007/s00415-009-5406-x.

[10] S. Nannoni *et al.*, “Correlation between ASPECTS and core volume on CT perfusion: Impact of time since stroke onset and presence of large-vessel occlusion,” in *American Journal of Neuroradiology*, American Society of Neuroradiology, Mar. 2021, pp. 422–428. doi: 10.3174/AJNR.A6959.

[11] W. Liang *et al.*, “Post-ASPECTS and Post-PC-ASPECTS Predict the Outcome of Anterior and Posterior Ischemic Stroke Following Thrombectomy,” *Risk Manag Healthc Policy*, vol. 16, pp. 2757–2769, 2023, doi: 10.2147/RMHP.S436661.

[12] S. B. Coutts *et al.*, “ASPECTS on CTA source images versus unenhanced CT: Added value in predicting final infarct extent and clinical outcome,” in *Stroke*, Nov. 2004, pp. 2472–2476. doi: 10.1161/01.STR.0000145330.14928.2a.

[13] S. F. Zaidi *et al.*, “Final infarct volume is a stronger predictor of outcome than recanalization in patients with proximal middle cerebral artery occlusion treated with endovascular therapy,” *Stroke*, vol. 43, no. 12, pp. 3238–3244, Dec. 2012, doi: 10.1161/STROKEAHA.112.671594.

[14] L. A. Beslow *et al.*, “Modified pediatric ASPECTS correlates with infarct volume in childhood arterial ischemic stroke,” *Front Neurol*, vol. JUL, 2012, doi: 10.3389/fneur.2012.00122.

[15] R. Dorent *et al.*, “Learning joint segmentation of tissues and brain lesions from task-specific hetero-modal domain-shifted datasets,” *Med Image Anal*, vol. 67, Jan. 2021, doi: 10.1016/j.media.2020.101862.

[16] G. Danala *et al.*, “Developing new quantitative CT image markers to predict prognosis of acute ischemic stroke patients,” *J Xray Sci Technol*, vol. 30, no. 3, pp. 459–475, 2022, doi: 10.3233/XST-221138.

[17] G. Danala *et al.*, “Applying Quantitative Radiographic Image Markers to Predict Clinical Complications After Aneurysmal Subarachnoid Hemorrhage: A Pilot Study,” *Ann Biomed Eng*, vol. 50, no. 4, pp. 413–425, Apr. 2022, doi: 10.1007/s10439-022-02926-z.

[18] P. Papanagiotou, W. Reith, A. Kastrup, and C. Roth, “Current reperfusion strategies for acute stroke,” Jan. 2014. doi: 10.1016/j.iccl.2013.09.008.

[19] A. Yafasova *et al.*, “Time to Thrombolysis and Long-Term Outcomes in Patients With Acute Ischemic Stroke: A Nationwide Study,” *Stroke*, vol. 52, no. 5, pp. 1724–1732, May 2021, doi: 10.1161/STROKEAHA.120.032837.

[20] S. Mirniaharikandehei, M. Heidari, G. Danala, S. Lakshmivarahan, and B. Zheng, “Applying a random projection algorithm to optimize machine learning model for predicting peritoneal metastasis in gastric cancer patients using CT images,” *Comput Methods Programs Biomed*, vol. 200, Mar. 2021, doi: 10.1016/j.cmpb.2021.105937.

[21] S. Yan, W. Qian, Y. Guan, and B. Zheng, “Improving lung cancer prognosis assessment by incorporating synthetic minority oversampling technique and score fusion method,” *Med Phys*, vol. 43, no. 6, pp. 2694–2703, Jun. 2016, doi: 10.1118/1.4948499.

[22] S. M. Seyedsaadat *et al.*, “Differential contribution of ASPECTS regions to clinical outcome after thrombectomy for acute ischemic stroke,” *American Journal of Neuroradiology*, vol. 42, no. 6, pp. 1104–1108, Jun. 2021, doi: 10.3174/ajnr.A7096.

[23] G. Brugnara *et al.*, “Clinical value of automated volumetric quantification of early ischemic tissue changes on non-contrast CT,” *J Neurointerv Surg*, vol. 15, no. e2, pp. E178–E183, Nov. 2023, doi: 10.1136/jnis-2022-019400.